

KINGSTONE, Australia

WMO: 958070

Lat: 35.7114S

Lon: 137.5231E

Elev: 22

StdP: 14.68

Time Zone: 9.50 (AUA)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.7	37.0	32.8	27.3	42.7	34.9	29.7	43.6	26.6	54.7	24.3	55.2	1.7	260	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.6	93.3	65.8	88.4	65.3	83.6	64.3	69.3	83.0	67.7	81.6	66.1	78.9	13.2	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	94.4	72.2	63.2	86.9	70.5	61.6	81.9	69.5	33.5	82.5	32.0	81.9	30.8	79.4	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.6	22.0	20.0	DB	31.1	103.5	1.4	4.1	30.0	106.5	29.2	108.9	28.4	111.2	27.3	114.2	(4)
(5)			WB	30.6	72.5	1.5	2.2	29.5	74.1	28.6	75.4	27.7	76.6	26.6	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	67.2	67.2	63.7	59.3	55.3	52.1	51.0	51.1	53.1	55.9	60.5	63.7	(6)	
	DBStd	7.53	5.88	5.56	5.31	4.41	3.95	3.60	3.57	3.57	4.16	4.87	5.83	5.70	(7)	
	HDD50	95	0	0	0	0	4	19	31	28	11	2	0	0	(8)	
	HDD65	2799	36	31	90	181	303	388	435	432	358	288	163	94	(9)	
	CDD50	3117	534	480	425	279	167	81	61	61	104	185	315	425	(10)	
	CDD65	345	105	92	49	10	1	0	0	0	1	5	28	54	(11)	
	CDH74	4229	1216	907	568	136	5	0	0	0	3	137	487	770	(12)	
	CDH80	1833	592	411	229	26	0	0	0	0	0	37	194	344	(13)	
(14) Wind	WSAvg	9.4	12.0	11.3	9.8	7.9	7.8	7.7	8.7	8.6	9.0	9.5	10.2	10.8	(14)	
(15) Precipitation	PrecAvg	19.1	0.6	0.8	0.8	1.2	2.0	3.0	2.9	2.5	2.0	1.3	0.9	0.9	(15)	
	PrecMax	27.3	2.5	2.8	4.2	2.7	3.3	5.2	5.2	3.9	5.0	4.0	2.4	2.4	(16)	
	PrecMin	12.9	0.1	0.0	0.0	0.1	0.4	0.9	1.1	0.6	0.6	0.0	0.2	0.2	(17)	
	PrecStd	3.7	0.6	0.8	0.9	0.7	0.7	1.1	1.0	1.0	1.1	1.0	0.6	0.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	101.8	98.1	93.3	83.6	73.7	65.0	63.4	67.2	73.1	85.3	92.2	97.4	(19)	
		MCWB	67.9	68.6	65.7	62.4	59.2	57.4	55.5	56.7	58.3	61.1	65.0	65.1	(20)	
	2%	DB	93.1	90.9	86.5	77.4	69.2	62.6	61.1	63.9	68.7	77.4	85.8	89.0	(21)	
		MCWB	66.4	66.0	65.0	61.2	59.3	56.2	54.5	55.2	58.0	59.9	63.3	63.7	(22)	
	5%	DB	86.5	84.8	80.1	73.8	66.2	61.0	59.7	61.4	65.4	71.8	79.6	82.5	(23)	
		MCWB	65.9	66.1	64.0	61.0	58.3	55.2	53.8	54.2	56.5	59.3	62.5	63.3	(24)	
	10%	DB	79.9	78.7	74.5	70.2	64.0	59.5	58.4	59.5	62.9	67.4	73.6	76.4	(25)	
		MCWB	64.8	64.7	62.8	60.0	57.5	54.3	53.0	53.3	55.5	57.8	61.0	62.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.8	72.5	69.8	66.1	63.3	59.4	57.6	58.6	61.4	64.1	68.1	69.0	(27)	
		MCDB	88.4	86.2	82.6	73.7	67.3	62.4	60.8	63.3	68.4	75.6	85.1	81.5	(28)	
	2%	WB	69.2	69.3	67.4	64.0	61.0	57.5	55.8	56.5	59.4	61.9	65.4	66.6	(29)	
		MCDB	85.3	80.7	79.3	72.4	65.8	61.1	59.5	61.5	65.7	72.8	80.3	81.4	(30)	
	5%	WB	67.2	67.5	65.5	62.5	59.6	56.2	54.7	55.1	57.7	60.1	63.5	64.6	(31)	
		MCDB	82.1	80.5	76.5	70.7	64.5	59.9	58.5	60.2	63.9	69.7	76.6	78.8	(32)	
	10%	WB	65.2	65.6	63.7	61.0	58.2	55.0	53.6	53.9	56.1	58.4	61.3	62.8	(33)	
		MCDB	78.6	76.9	73.1	68.5	63.2	58.8	57.5	58.7	61.8	66.5	72.2	75.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.6	20.3	20.8	21.0	17.0	14.8	14.5	16.5	18.5	21.6	22.0	22.6	(35)	
		MCDBR	36.3	33.3	32.2	28.8	21.5	16.5	15.0	19.3	23.5	32.3	34.7	36.0	(36)	
		MCWBR	12.6	11.8	13.7	15.0	13.6	11.6	10.3	12.8	14.4	17.1	14.8	13.6	(37)	
	5% WB	MCDBR	28.7	26.0	25.7	23.8	18.1	15.3	14.4	18.1	20.7	27.3	29.3	28.1	(38)	
		MCWBR	11.6	10.4	12.5	13.5	12.3	11.3	10.6	12.8	13.4	15.6	13.7	12.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.344	0.342	0.324	0.319	0.312	0.306	0.305	0.314	0.338	0.344	0.343	0.339	(40)	
	taud		2.564	2.582	2.631	2.628	2.625	2.626	2.622	2.583	2.508	2.492	2.523	2.561	(41)	
	Ebn at Noon		315	308	303	285	270	262	269	283	292	304	312	318	(42)	
	Edh at Noon		34	32	28	25	22	21	22	26	31	35	35	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		2321	1990	1601	1157	799	660	718	994	1381	1775	2122	2278	(44)	
	RadStd		112	93	61	72	37	37	35	58	107	121	100	103	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+232	+95
(47) Regional (23 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-176	+290	+126

Nomenclature: See separate page